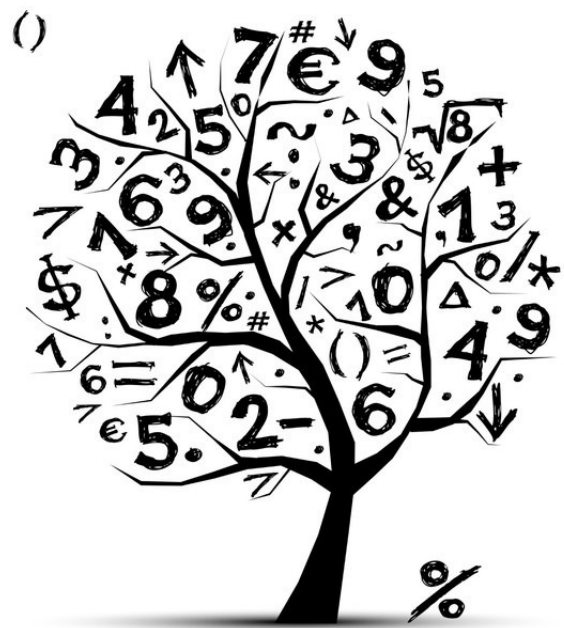


Last update: October 7, 2026

Inversion of matrices



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1 Properties of the inverse matrix

- Recall (see the chapter on matrices) that a square matrix $A \in \mathbb{R}^{n \times n}$ is **invertible** if there exists a matrix $A^{-1} \in \mathbb{R}^{n \times n}$ such that $AA^{-1} = A^{-1}A = I$. A square matrix which is not invertible is called **singular**.
- In this chapter, A' denotes the transpose of A , as in the chapter on matrices.

Proposition 1: Uniqueness of the inverse

Let $A \in \mathbb{R}^{n \times n}$ be invertible. If $B, C \in \mathbb{R}^{n \times n}$ are such that $AB = BA = I$ and $AC = CA = I$, then $B = C$.

Proof. Using the associativity of the matrix product, we have:

$$B = BI = B(AC) = (BA)C = IC = C.$$

□

Proposition 2: Properties of the inverse

Let $A, B \in \mathbb{R}^{n \times n}$ be invertible matrices. Then:

- A^{-1} is invertible and $(A^{-1})^{-1} = A$;
- AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$;
- A' is invertible and $(A')^{-1} = (A^{-1})'$;
- $\det(A) \neq 0$ and $\det(A^{-1}) = \frac{1}{\det(A)}$.

Proof. By the uniqueness of the inverse, in each case it is enough to exhibit a matrix that multiplied on the left and on the right gives I .

1. From $A^{-1}A = AA^{-1} = I$, the matrix A is the inverse of A^{-1} .

2. Using associativity:

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I,$$

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}IB = B^{-1}B = I.$$

3. Recall that $(CD)' = D'C'$ for all $C, D \in \mathbb{R}^{n \times n}$: indeed, the entry (i, j) of $(CD)'$ is the entry (j, i) of CD , i.e., $\sum_{k=1}^n c_{jk}d_{ki} = \sum_{k=1}^n d'_{ik}c'_{kj}$, which is the entry (i, j) of $D'C'$. Then, since $I' = I$:

$$A'(A^{-1})' = (A^{-1}A)' = I' = I, \quad (A^{-1})'A' = (AA^{-1})' = I' = I.$$

4. From the product rule of determinants:

$$\det(A)\det(A^{-1}) = \det(AA^{-1}) = \det(I) = 1.$$

Hence $\det(A) \neq 0$ and $\det(A^{-1}) = 1/\det(A)$.

□

- Note the reversed order in $(AB)^{-1} = B^{-1}A^{-1}$: since the matrix product is not commutative, in general $(AB)^{-1} \neq A^{-1}B^{-1}$.
- Property 4 shows that a matrix with $\det(A) = 0$ cannot be invertible, i.e., it is singular.

2 Computation of the inverse

2.1 Gauss–Jordan method

Gauss–Jordan method (idea). To compute A^{-1} , we build the augmented matrix

$$(A \mid I)$$

and we apply **elementary row operations** to transform the left block into the identity matrix. If we obtain

$$(I \mid B),$$

then $B = A^{-1}$.

Example 1: inverse matrix via Gauss–Jordan elimination

Let us consider

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

We compute the inverse matrix A^{-1} with the method of Gauss–Jordan.

$$\left(\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{array} \right)$$

$$\left(\begin{array}{cc|cc} 1 & 1 & 0 & 1 \\ 2 & 1 & 1 & 0 \end{array} \right) \quad (R_1 \leftrightarrow R_2)$$

$$\left(\begin{array}{cc|cc} 1 & 1 & 0 & 1 \\ 0 & -1 & 1 & -2 \end{array} \right) \quad (R_2 \leftarrow R_2 - 2R_1)$$

$$\left(\begin{array}{cc|cc} 1 & 1 & 0 & 1 \\ 0 & 1 & -1 & 2 \end{array} \right) \quad (R_2 \leftarrow -R_2)$$

$$\left(\begin{array}{cc|cc} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 2 \end{array} \right) \quad (R_1 \leftarrow R_1 - R_2)$$

Since the left block is the identity matrix, we obtain:

$$A^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}.$$

We can verify:

$$AA^{-1} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I.$$

Example 2: inverse matrix via Gauss–Jordan elimination

Let us consider

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{pmatrix}.$$

We compute the inverse matrix A^{-1} with the method of Gauss–Jordan.

$$\begin{pmatrix} 1 & 2 & 0 & | & 1 & 0 & 0 \\ 0 & 1 & 1 & | & 0 & 1 & 0 \\ 2 & 0 & 1 & | & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 0 & | & 1 & 0 & 0 \\ 0 & 1 & 1 & | & 0 & 1 & 0 \\ 0 & -4 & 1 & | & -2 & 0 & 1 \end{pmatrix} \quad (R_3 \leftarrow R_3 - 2R_1)$$

$$\begin{pmatrix} 1 & 0 & -2 & | & 1 & -2 & 0 \\ 0 & 1 & 1 & | & 0 & 1 & 0 \\ 0 & -4 & 1 & | & -2 & 0 & 1 \end{pmatrix} \quad (R_1 \leftarrow R_1 - 2R_2)$$

$$\begin{pmatrix} 1 & 0 & -2 & | & 1 & -2 & 0 \\ 0 & 1 & 1 & | & 0 & 1 & 0 \\ 0 & 0 & 5 & | & -2 & 4 & 1 \end{pmatrix} \quad (R_3 \leftarrow R_3 + 4R_2)$$

$$\begin{pmatrix} 1 & 0 & -2 & | & 1 & -2 & 0 \\ 0 & 1 & 1 & | & 0 & 1 & 0 \\ 0 & 0 & 1 & | & -\frac{2}{5} & \frac{4}{5} & \frac{1}{5} \end{pmatrix} \quad (R_3 \leftarrow \frac{1}{5}R_3)$$

$$\begin{pmatrix} 1 & 0 & 0 & | & \frac{1}{5} & -\frac{2}{5} & \frac{2}{5} \\ 0 & 1 & 1 & | & 0 & 1 & 0 \\ 0 & 0 & 1 & | & -\frac{2}{5} & \frac{4}{5} & \frac{1}{5} \end{pmatrix} \quad (R_1 \leftarrow R_1 + 2R_3)$$

$$\begin{pmatrix} 1 & 0 & 0 & | & \frac{1}{5} & -\frac{2}{5} & \frac{2}{5} \\ 0 & 1 & 0 & | & \frac{2}{5} & \frac{1}{5} & -\frac{1}{5} \\ 0 & 0 & 1 & | & -\frac{2}{5} & \frac{4}{5} & \frac{1}{5} \end{pmatrix} \quad (R_2 \leftarrow R_2 - R_3)$$

Since the left block is the identity matrix, we obtain:

$$A^{-1} = \begin{pmatrix} \frac{1}{5} & -\frac{2}{5} & \frac{2}{5} \\ \frac{2}{5} & \frac{1}{5} & -\frac{1}{5} \\ -\frac{2}{5} & \frac{4}{5} & \frac{1}{5} \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 1 & -2 & 2 \\ 2 & 1 & -1 \\ -2 & 4 & 1 \end{pmatrix}.$$

Example 3: inverse matrix via Gauss–Jordan elimination with a row swap

Let us consider

$$A = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 4 & -3 & 8 \end{pmatrix}.$$

We compute the inverse matrix A^{-1} with the method of Gauss–Jordan. Since $a_{11} = 0$, we start with a row swap.

$$\begin{pmatrix} 0 & 1 & 2 & | & 1 & 0 & 0 \\ 1 & 0 & 3 & | & 0 & 1 & 0 \\ 4 & -3 & 8 & | & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 3 & | & 0 & 1 & 0 \\ 0 & 1 & 2 & | & 1 & 0 & 0 \\ 4 & -3 & 8 & | & 0 & 0 & 1 \end{pmatrix} \quad (R_1 \leftrightarrow R_2)$$

$$\begin{pmatrix} 1 & 0 & 3 & | & 0 & 1 & 0 \\ 0 & 1 & 2 & | & 1 & 0 & 0 \\ 0 & -3 & -4 & | & 0 & -4 & 1 \end{pmatrix} \quad (R_3 \leftarrow R_3 - 4R_1)$$

$$\begin{pmatrix} 1 & 0 & 3 & | & 0 & 1 & 0 \\ 0 & 1 & 2 & | & 1 & 0 & 0 \\ 0 & 0 & 2 & | & 3 & -4 & 1 \end{pmatrix} \quad (R_3 \leftarrow R_3 + 3R_2)$$

$$\begin{pmatrix} 1 & 0 & 3 & | & 0 & 1 & 0 \\ 0 & 1 & 2 & | & 1 & 0 & 0 \\ 0 & 0 & 1 & | & \frac{3}{2} & -2 & \frac{1}{2} \end{pmatrix} \quad (R_3 \leftarrow \frac{1}{2}R_3)$$

$$\begin{pmatrix} 1 & 0 & 0 & | & -\frac{9}{2} & 7 & -\frac{3}{2} \\ 0 & 1 & 2 & | & 1 & 0 & 0 \\ 0 & 0 & 1 & | & \frac{3}{2} & -2 & \frac{1}{2} \end{pmatrix} \quad (R_1 \leftarrow R_1 - 3R_3)$$

$$\begin{pmatrix} 1 & 0 & 0 & | & -\frac{9}{2} & 7 & -\frac{3}{2} \\ 0 & 1 & 0 & | & -2 & 4 & -1 \\ 0 & 0 & 1 & | & \frac{3}{2} & -2 & \frac{1}{2} \end{pmatrix} \quad (R_2 \leftarrow R_2 - 2R_3)$$

Since the left block is the identity matrix, we obtain:

$$A^{-1} = \begin{pmatrix} -\frac{9}{2} & 7 & -\frac{3}{2} \\ -2 & 4 & -1 \\ \frac{3}{2} & -2 & \frac{1}{2} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} -9 & 14 & -3 \\ -4 & 8 & -2 \\ 3 & -4 & 1 \end{pmatrix}.$$

Observation 1: Singular matrices and the Gauss–Jordan method

Let $A \in \mathbb{R}^{n \times n}$. If, while applying elementary row operations to $(A \mid I)$, a **zero row appears in the left block**, then $\det(A) = 0$ and A is singular: the method stops and A^{-1} does not exist.

Proof. The left block is obtained from A by elementary row operations. Each of them multiplies the determinant by a nonzero number (-1 for a row swap, $\lambda \neq 0$ for a row scaling, 1 for a row addition). A matrix with a zero row has zero determinant (Laplace expansion along that row), hence $\det(A) = 0$. By property 4 of Proposition 2, A is not invertible. $\square$

Example 4: Gauss–Jordan elimination on a singular matrix

Let us consider

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ 1 & 3 & 4 \end{pmatrix}.$$

We try to compute A^{-1} with the method of Gauss–Jordan.

$$\begin{pmatrix} 1 & 2 & 3 & | & 1 & 0 & 0 \\ 2 & 5 & 7 & | & 0 & 1 & 0 \\ 1 & 3 & 4 & | & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 3 & | & 1 & 0 & 0 \\ 0 & 1 & 1 & | & -2 & 1 & 0 \\ 1 & 3 & 4 & | & 0 & 0 & 1 \end{pmatrix} \quad (R_2 \leftarrow R_2 - 2R_1)$$

$$\begin{pmatrix} 1 & 2 & 3 & | & 1 & 0 & 0 \\ 0 & 1 & 1 & | & -2 & 1 & 0 \\ 0 & 1 & 1 & | & -1 & 0 & 1 \end{pmatrix} \quad (R_3 \leftarrow R_3 - R_1)$$

$$\begin{pmatrix} 1 & 2 & 3 & | & 1 & 0 & 0 \\ 0 & 1 & 1 & | & -2 & 1 & 0 \\ 0 & 0 & 0 & | & 1 & -1 & 1 \end{pmatrix} \quad (R_3 \leftarrow R_3 - R_2)$$

The third row of the left block is zero: the left block cannot be transformed into the identity matrix, and the method stops. Hence A is singular. Indeed, the third row of A is the difference of the second and the first row, and

$$\det(A) = 1 \cdot (5 \cdot 4 - 7 \cdot 3) - 2 \cdot (2 \cdot 4 - 7 \cdot 1) + 3 \cdot (2 \cdot 3 - 5 \cdot 1) = -1 - 2 + 3 = 0.$$

2.2 Inverse via the adjugate matrix

- Recall (see the chapter on matrices) that, given $A \in \mathbb{R}^{n \times n}$, the **cofactor** of the entry a_{ij} is $C_{ij} = (-1)^{i+j} \det(A_{ij})$, where A_{ij} is the minor matrix obtained from A by deleting row i and column j .

Definition 1: Cofactor matrix and adjugate matrix

Let $A \in \mathbb{R}^{n \times n}$ with $n \geq 2$. The **cofactor matrix** of A is the matrix $C = (C_{ij}) \in \mathbb{R}^{n \times n}$ of the cofactors of A . The **adjugate matrix** of A is the transpose of the cofactor matrix:

$$\text{adj}(A) = C', \quad \text{i.e.,} \quad (\text{adj}(A))_{ij} = C_{ji}.$$

Theorem 1: Adjugate formula

Let $A \in \mathbb{R}^{n \times n}$ with $n \geq 2$. Then

$$A \text{adj}(A) = \text{adj}(A) A = \det(A) I.$$

Consequently, if $\det(A) \neq 0$, then A is invertible and

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A).$$

Proof. The entry (i, j) of $A \text{adj}(A)$ is

$$\sum_{k=1}^n a_{ik} C_{jk}.$$

- If $i = j$, this is the Laplace expansion of $\det(A)$ along row i .
- If $i \neq j$, this is the Laplace expansion along row j of the matrix $\tilde{A}$ obtained from A by replacing row j with row i (the cofactors C_{jk} do not depend on row j). The matrix $\tilde{A}$ has two equal rows: swapping them leaves $\tilde{A}$ unchanged and changes the sign of its determinant, so $\det(\tilde{A}) = -\det(\tilde{A})$, i.e., $\det(\tilde{A}) = 0$.

Hence $A \text{adj}(A) = \det(A) I$. The identity $\text{adj}(A) A = \det(A) I$ is obtained in the same way, using Laplace expansions along columns. If $\det(A) \neq 0$, dividing by $\det(A)$ shows that $\frac{1}{\det(A)} \text{adj}(A)$ is the inverse of A . $\square$

Observation 2: Invertibility and determinant

A square matrix $A \in \mathbb{R}^{n \times n}$ is invertible if and only if $\det(A) \neq 0$.

Proof. If A is invertible, then $\det(A) \neq 0$ by property 4 of Proposition 2. Conversely, if $\det(A) \neq 0$, then A is invertible by Theorem 1 (for $n = 1$, $A = (a)$ with $a \neq 0$ and $A^{-1} = (1/a)$). $\square$

Observation 3: Inverse of a 2×2 matrix

Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathbb{R}^{2 \times 2} \quad \text{with} \quad \det(A) = ad - bc \neq 0.$$

Then

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Proof. The minor matrices of A are 1×1 , so the cofactors are $C_{11} = d$, $C_{12} = -c$, $C_{21} = -b$, $C_{22} = a$. Hence

$$C = \begin{pmatrix} d & -c \\ -b & a \end{pmatrix}, \quad \text{adj}(A) = C' = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix},$$

and the formula follows from Theorem 1. $\square$

Inverse of a 2×2 matrix (rule). Swap the two diagonal entries, change the sign of the two off-diagonal entries, and divide by the determinant.

Example 5: inverse of a 2×2 matrix with the formula

Let us consider again

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

We have $\det(A) = 2 \cdot 1 - 1 \cdot 1 = 1 \neq 0$, hence

$$A^{-1} = \frac{1}{1} \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix},$$

which is the matrix obtained with the method of Gauss–Jordan in Example 1.

Example 6: inverse of a 3×3 matrix with the adjugate formula

Let us consider again

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{pmatrix}.$$

By Laplace expansion along the first row:

$$\det(A) = 1 \cdot (1 \cdot 1 - 1 \cdot 0) - 2 \cdot (0 \cdot 1 - 1 \cdot 2) + 0 = 1 + 4 = 5 \neq 0.$$

The cofactors are:

$$\begin{aligned} C_{11} &= +\det \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = 1, & C_{12} &= -\det \begin{pmatrix} 0 & 1 \\ 2 & 1 \end{pmatrix} = 2, & C_{13} &= +\det \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix} = -2, \\ C_{21} &= -\det \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} = -2, & C_{22} &= +\det \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} = 1, & C_{23} &= -\det \begin{pmatrix} 1 & 2 \\ 2 & 0 \end{pmatrix} = 4, \\ C_{31} &= +\det \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix} = 2, & C_{32} &= -\det \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = -1, & C_{33} &= +\det \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = 1. \end{aligned}$$

Hence

$$C = \begin{pmatrix} 1 & 2 & -2 \\ -2 & 1 & 4 \\ 2 & -1 & 1 \end{pmatrix}, \quad \text{adj}(A) = C' = \begin{pmatrix} 1 & -2 & 2 \\ 2 & 1 & -1 \\ -2 & 4 & 1 \end{pmatrix},$$

and

$$A^{-1} = \frac{1}{5} \begin{pmatrix} 1 & -2 & 2 \\ 2 & 1 & -1 \\ -2 & 4 & 1 \end{pmatrix},$$

which is the matrix obtained with the method of Gauss–Jordan in Example 2.

- For $n = 2$ (and often for $n = 3$) the adjugate formula is convenient for computations by hand. For larger matrices, the method of Gauss–Jordan requires far fewer operations.